ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 150°	One part $= 360^\circ \div 12 = 30^\circ$.
2	**(a)** 55°	Let angles be x and $x + d$; they add to 180°.
3	**(c)** 480 cm^2	Other side $= 30$ cm; area $= 16 \times 30$.
4	**(c)** 52 cm	Side $= 13$ cm, perimeter $= 4 \times 13$.
5	**(c)** 120 cm^2	Area $= \frac{1}{2} d_1 d_2$.
6	**(a)** 43 cm	Each side of the new figure is half a diagonal; there are two of each.
7	**(b)** 44	Opposite angles are equal.
8	**(b)** 60 cm	Midpoint theorem: $BC = 2DE$.
9	**(a)** 144°	One part $= 360^\circ \div 15 = 24^\circ$.
10	**(d)** 50°	Let angles be x and $x + d$; they add to 180°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let the angles be x and $x + 20$; adjacent angles sum to 180°; $2x + 20 = 180$ gives $x = 80$ (1); Angles: $80^\circ, 100^\circ, 80^\circ, 100^\circ$ (1).	1+1
12	Other side $= \sqrt{34^2 - 16^2} = 30$ cm; Area $= 16 \times 30 = 480$ cm^2 (1); Perimeter $= 2(16 + 30) = 92$ cm (1).	1+1
13	Diagonals bisect each other at right angles; Side $= \sqrt{7^2 + 24^2} = 25$ cm (1); Perimeter $= 4 \times 25 = 100$ cm; Area $= \frac{1}{2} \times 14 \times 48 = 336$ cm^2 (1).	1+1
14	Let the angles be x and $x + 40$; adjacent angles sum to 180°; $2x + 40 = 180$ gives $x = 70$ (1); Angles: $70^\circ, 110^\circ, 70^\circ, 110^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= \frac{1}{2} d^2 = \frac{1}{2} \times 18^2 = 162$ cm^2 (1); Side$^2 = 162$ (1); Side $= 9\sqrt{2}$ cm (1).	1+1+1
16	$\angle B = 180^\circ - 64^\circ = 116^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 32^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 58^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 32^\circ - 58^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
17	Opposite sides are equal: $AB = CD$; $2x + 7 = x + 15$ gives $x = 8$ (1); $AB = CD = 23$ cm and $BC = AD = 14$ cm (1); Perimeter $= 2(23 + 14) = 74$ cm (1).	1+1+1
	OR: Area $= \frac{1}{2} d^2 = \frac{1}{2} \times 34^2 = 578$ cm^2 (1); Side$^2 = 578$ (1); Side $= 17\sqrt{2}$ cm (1).	1+1+1
18	$\angle B = 180^\circ - 98^\circ = 82^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 49^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 41^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 49^\circ - 41^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
	OR: Opposite sides are equal: $AB = CD$; $2x + 1 = x + 8$ gives $x = 7$ (1); $AB = CD = 15$ cm and $BC = AD = 9$ cm (1); Perimeter $= 2(15 + 9) = 48$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	In $\triangle ABC$, $AB^2 + BC^2 = 121 + 36 = 157$ (as $BC = AD = 6$) (1); This equals AC^2, so $\angle B = 90^\circ$ (converse of the Baudhāyana–Pythagoras theorem) (1); A parallelogram with one right angle has all angles 90° (1); Hence ABCD is a rectangle (2).	1+1+1+2
	OR: Midpoint theorem on each pair of sides (1); $DE = \frac{1}{2} AC$, $EF = \frac{1}{2} AB$, $FD = \frac{1}{2} BC$ (2); Perimeter of $\triangle DEF = \frac{1}{2}(82) = 41$ cm (2).	1+2+2
20	In $\triangle ABC$, $AB^2 + BC^2 = 169 + 36 = 205$ (as $BC = AD = 6$) (1); This equals AC^2, so $\angle B = 90^\circ$ (converse of the Baudhāyana–Pythagoras theorem) (1); A parallelogram with one right angle has all angles 90° (1); Hence ABCD is a rectangle (2).	1+1+1+2
	OR: Midpoint theorem on each pair of sides (1); $DE = \frac{1}{2} AC$, $EF = \frac{1}{2} AB$, $FD = \frac{1}{2} BC$ (2); Perimeter of $\triangle DEF = \frac{1}{2}(56) = 28$ cm (2).	1+2+2

Mistakes to avoid:

- Saying the diagonals of every parallelogram are equal; that holds only for a rectangle (and square).
- Saying the diagonals of a rhombus are equal; they are perpendicular bisectors of each other, equal only in a square.
- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
- In the midpoint theorem, writing $DE = 2BC$ or $DE = \frac{1}{2}AB$; the segment is half the *third* side that it is parallel to.
- Forgetting that a quadrilateral's angles must add up to 360° (not 180°) when solving for x.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 60 cm^2	Other side $= 12$ cm; area $= 5 \times 12$.
2	**(d)** 100 cm	Side $= 25$ cm, perimeter $= 4 \times 25$.
3	**(a)** 2160 cm^2	Area $= \frac{1}{2}d_1d_2$.
4	**(b)** 50 cm	Each side of the new figure is half a diagonal; there are two of each.
5	**(a)** 55	Opposite angles are equal.
6	**(b)** 38 cm	Midpoint theorem: $BC = 2DE$.
7	**(c)** 144°	One part $= 360^\circ \div 15 = 24^\circ$.
8	**(a)** 70°	Let angles be x and $x + d$; they add to 180°.
9	**(d)** 540 cm^2	Other side $= 36$ cm; area $= 15 \times 36$.
10	**(a)** 156 cm	Side $= 39$ cm, perimeter $= 4 \times 39$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Other side $= \sqrt{25^2 - 7^2} = 24$ cm; Area $= 7 \times 24 = 168$ cm^2 (1); Perimeter $= 2(7 + 24) = 62$ cm (1).	1+1
12	Diagonals bisect each other at right angles; Side $= \sqrt{14^2 + 48^2} = 50$ cm (1); Perimeter $= 4 \times 50 = 200$ cm; Area $= \frac{1}{2} \times 28 \times 96 = 1344$ cm^2 (1).	1+1
13	Let the angles be x and $x + 50$; adjacent angles sum to 180°; $2x + 50 = 180$ gives $x = 65$ (1); Angles: $65^\circ, 115^\circ, 65^\circ, 115^\circ$ (1).	1+1
14	Other side $= \sqrt{26^2 - 10^2} = 24$ cm; Area $= 10 \times 24 = 240$ cm^2 (1); Perimeter $= 2(10 + 24) = 68$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= \frac{1}{2}d^2 = \frac{1}{2} \times 38^2 = 722$ cm^2 (1); Side$^2 = 722$ (1); Side $= 19\sqrt{2}$ cm (1).	1+1+1
16	$\angle B = 180^\circ - 54^\circ = 126^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 27^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 63^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 27^\circ - 63^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
17	Opposite sides are equal: $AB = CD$; $2x + 3 = x + 9$ gives $x = 6$ (1); $AB = CD = 15$ cm and $BC = AD = 11$ cm (1); Perimeter $= 2(15 + 11) = 52$ cm (1).	1+1+1
	OR: Area $= \frac{1}{2}d^2 = \frac{1}{2} \times 8^2 = 32$ cm^2 (1); Side$^2 = 32$ (1); Side $= 4\sqrt{2}$ cm (1).	1+1+1
18	$\angle B = 180^\circ - 62^\circ = 118^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 31^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 59^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 31^\circ - 59^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
	OR: Opposite sides are equal: $AB = CD$; $2x + 1 = x + 11$ gives $x = 10$ (1); $AB = CD = 21$ cm and $BC = AD = 12$ cm (1); Perimeter $= 2(21 + 12) = 66$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	In $\triangle ABC$, $AB^2 + BC^2 = 169 + 196 = 365$ (as $BC = AD = 14$) (1); This equals AC^2, so $\angle B = 90^\circ$ (converse of the Baudhāyana–Pythagoras theorem) (1); A parallelogram with one right angle has all angles 90° (1); Hence ABCD is a rectangle (2).	1+1+1+2
	OR: Midpoint theorem on each pair of sides (1); $DE = \frac{1}{2}AC$, $EF = \frac{1}{2}AB$, $FD = \frac{1}{2}BC$ (2); Perimeter of $\triangle DEF = \frac{1}{2}(116) = 58$ cm (2).	1+2+2
20	In $\triangle ABC$, $AB^2 + BC^2 = 49 + 169 = 218$ (as $BC = AD = 13$) (1); This equals AC^2, so $\angle B = 90^\circ$ (converse of the Baudhāyana–Pythagoras theorem) (1); A parallelogram with one right angle has all angles 90° (1); Hence ABCD is a rectangle (2).	1+1+1+2
	OR: Midpoint theorem on each pair of sides (1); $DE = \frac{1}{2}AC$, $EF = \frac{1}{2}AB$, $FD = \frac{1}{2}BC$ (2); Perimeter of $\triangle DEF = \frac{1}{2}(84) = 42$ cm (2).	1+2+2

Mistakes to avoid:

- Saying the diagonals of every parallelogram are equal; that holds only for a rectangle (and square).
- Saying the diagonals of a rhombus are equal; they are perpendicular bisectors of each other, equal only in a square.
- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
- In the midpoint theorem, writing $DE = 2BC$ or $DE = \frac{1}{2}AB$; the segment is half the *third* side that it is parallel to.
- Forgetting that a quadrilateral's angles must add up to 360° (not 180°) when solving for x.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 240 cm^2	Area $= \frac{1}{2}d_1d_2$.
2	**(d)** 61 cm	Each side of the new figure is half a diagonal; there are two of each.
3	**(d)** 54	Opposite angles are equal.
4	**(b)** 46 cm	Midpoint theorem: $BC = 2DE$.
5	**(d)** 144°	One part $= 360^\circ \div 10 = 36^\circ$.
6	**(d)** 65°	Let angles be x and $x + d$; they add to 180°.
7	**(a)** 240 cm^2	Other side $= 24$ cm; area $= 10 \times 24$.
8	**(b)** 200 cm	Side $= 50$ cm, perimeter $= 4 \times 50$.
9	**(a)** 1080 cm^2	Area $= \frac{1}{2}d_1d_2$.
10	**(b)** 45 cm	Each side of the new figure is half a diagonal; there are two of each.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Diagonals bisect each other at right angles; Side $= \sqrt{5^2 + 12^2} = 13$ cm (1); Perimeter $= 4 \times 13 = 52$ cm; Area $= \frac{1}{2} \times 10 \times 24 = 120$ cm^2 (1).	1+1
12	Let the angles be x and $x + 30$; adjacent angles sum to 180°; $2x + 30 = 180$ gives $x = 75$ (1); Angles: $75^\circ, 105^\circ, 75^\circ, 105^\circ$ (1).	1+1
13	Other side $= \sqrt{50^2 - 14^2} = 48$ cm; Area $= 14 \times 48 = 672$ cm^2 (1); Perimeter $= 2(14 + 48) = 124$ cm (1).	1+1
14	Diagonals bisect each other at right angles; Side $= \sqrt{8^2 + 15^2} = 17$ cm (1); Perimeter $= 4 \times 17 = 68$ cm; Area $= \frac{1}{2} \times 16 \times 30 = 240$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= \frac{1}{2}d^2 = \frac{1}{2} \times 20^2 = 200$ cm^2 (1); Side$^2 = 200$ (1); Side $= 10\sqrt{2}$ cm (1).	1+1+1
16	$\angle B = 180^\circ - 40^\circ = 140^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 20^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 70^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 20^\circ - 70^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
17	Opposite sides are equal: $AB = CD$; $3x + 9 = x + 19$ gives $x = 5$ (1); $AB = CD = 24$ cm and $BC = AD = 7$ cm (1); Perimeter $= 2(24 + 7) = 62$ cm (1).	1+1+1
	OR: Area $= \frac{1}{2}d^2 = \frac{1}{2} \times 16^2 = 128$ cm^2 (1); Side$^2 = 128$ (1); Side $= 8\sqrt{2}$ cm (1).	1+1+1
18	$\angle B = 180^\circ - 120^\circ = 60^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 60^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 30^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 60^\circ - 30^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
	OR: Opposite sides are equal: $AB = CD$; $3x + 7 = x + 29$ gives $x = 11$ (1); $AB = CD = 40$ cm and $BC = AD = 13$ cm (1); Perimeter $= 2(40 + 13) = 106$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	In $\triangle ABC$, $AB^2 + BC^2 = 196 + 81 = 277$ (as $BC = AD = 9$) (1); This equals AC^2, so $\angle B = 90^\circ$ (converse of the Baudhāyana–Pythagoras theorem) (1); A parallelogram with one right angle has all angles 90° (1); Hence ABCD is a rectangle (2).	1+1+1+2
	OR: Midpoint theorem on each pair of sides (1); $DE = \frac{1}{2}AC$, $EF = \frac{1}{2}AB$, $FD = \frac{1}{2}BC$ (2); Perimeter of $\triangle DEF = \frac{1}{2}(102) = 51$ cm (2).	1+2+2
20	In $\triangle ABC$, $AB^2 + BC^2 = 81 + 36 = 117$ (as $BC = AD = 6$) (1); This equals AC^2, so $\angle B = 90^\circ$ (converse of the Baudhāyana–Pythagoras theorem) (1); A parallelogram with one right angle has all angles 90° (1); Hence ABCD is a rectangle (2).	1+1+1+2
	OR: Midpoint theorem on each pair of sides (1); $DE = \frac{1}{2}AC$, $EF = \frac{1}{2}AB$, $FD = \frac{1}{2}BC$ (2); Perimeter of $\triangle DEF = \frac{1}{2}(72) = 36$ cm (2).	1+2+2

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- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
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